

Tutorial Sheet 1 (Limits, Continuity, Chain Rule, Change of Variables, and Jacobian)

1. Using the $\delta - \varepsilon$ approach, show that

a. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0$

c. $\lim_{(x,y) \rightarrow (1,1)} (x^2 + y^2 - 1) = 1$

b. $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) = 0$

2. Evaluate the following limits (if exist)

a. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2}$

e. $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \sin\left(\frac{1}{xy}\right)$

b. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2}}$

f. $\lim_{(x,y) \rightarrow (0,0)} \frac{x-2y}{x+y}$

c. $\lim_{(x,y) \rightarrow (0,0)} \frac{y^2 - x^2}{y^2 + x^2}$

g. $\lim_{(x,y) \rightarrow (0,0)} \frac{2y^4 x}{y^8 + 6x^2}$

d. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 y}{x^6 + y^2}$

3. In a robotic path planning the robot's position error is given by the function $f(x, y) = \frac{x^2 y - y^3 + \cos(xy)}{x^2 + y^2 + 1}$. Compute $\lim_{(x,y) \rightarrow (0,-1)} f(x, y)$.

4. Check whether the following functions are continuous or not

a. $f(x, y) = x^2 + y^2$ at $(1, 2)$

c. $f(x, y) = \begin{cases} \frac{xy}{x^2 + 5y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

b. $f(x, y) = \begin{cases} \frac{1}{1+e^x} + y^2, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

5. The electrical potential in a circuit is $V(x, y) = \frac{x^2 - y^2 + \sin(xy)}{x^2 + y^2 + 1}$. Check whether $V(x, y)$ is continuous at $(1, -1)$ or not.

6. If $z = x^3 y^2 + 3xy$, find the values of $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$, $\frac{\partial^2 z}{\partial x^2}$ and $\frac{\partial^2 z}{\partial y^2}$.

7. If $u = e^{xy+yz+zx}$, find the value of $\frac{\partial^3 u}{\partial x \partial y \partial z}$.

8. Compute $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$ for the function $f(x, y) = \begin{cases} \frac{x^3 y}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$

9. Find $\frac{\partial x}{\partial u}$ and $\frac{\partial y}{\partial u}$, where the equations $u = x^2 - y^2$ and $v = x^2 - y$ define x and y as functions of the independent variables u and v , the partial derivatives exist. Also, if $s = x^2 + y^2$, find $\frac{\partial s}{\partial u}$.

10. Let $f(x, y, z) = x^3 y^2 + yz^4$ and $x = s^2$, $y = st^2$, $z = s^2 t$. Using the chain rule, compute $\frac{\partial f}{\partial s}$.

11. If $u = \sec^{-1}\left(\frac{x^3 - y^3}{x - y}\right)$, then find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$.

12. Let $u = \frac{y^3 - x^3}{x^2 + y^2}$ prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = u$ and hence show that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 0$.

13. Use chain rule to find the derivative of $z = e^{xy}$ w.r.t t along the path $x = \cos t$, $y = \sin t$. Also find $\frac{dz}{dt}$ at $t = \frac{\pi}{2}$.

14. Suppose that a program's execution time depends on CPU cycles (x) and the memory usage (y) modeled by the equation $T(x, y) = x^2 + y^2$; where $x = p^2 - 1$ (CPU cycles depending on input parameter p) and $y = p^3 - p$ (memory usage depending on input parameter p). Use Chain rule of

partial differentiation to find the total derivative $\frac{dT}{dp}$; i.e., how execution time changes as we change the input parameter p .

15. Find the total differentiation of the following functions:

a. $z = \tan^{-1}\left(\frac{x}{y}\right)$; $(x, y) = (0, 0)$ b. $u = \left(xz + \frac{x}{z}\right)^y$; $z \neq 0$

16. Given transformation $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$ and $z = r \cos \theta$. Find the Jacobian.

17. In a computer graphics application, an image is transformed from original coordinates (x, y) to new coordinates (u, v) using the transformation: $u = 2x + y$ and $v = x - 2y$. Find the Jacobian.

Answers:

1.

2. a. 0 b. 0 e. 0

3. 1

4. a. Continuous b. Discontinuous c. Not continuous

5. Continuous

6. $\frac{\partial z}{\partial x} = 3x^2y^2 + 3y$, $\frac{\partial z}{\partial y} = 2x^3y + 3x$, $\frac{\partial^2 z}{\partial x^2} = 6xy^2$ and $\frac{\partial^2 z}{\partial y^2} = 2x^3$.

7. $e^{xy+yz+zx} [2(x+y+z) + (x+y)(x+z)(y+z)]$

8. $f_{xy}(0, 0) = 0$ and $f_{yx}(0, 0) = 0$

9. $\frac{\partial s}{\partial u} = \frac{1+2y}{1-2y}$

10. $\frac{\partial f}{\partial s} = 8s^7t^4 + 9s^8t^6$

11. $xu_x + yu_y = \frac{2(x^2+xy+y^2)}{|x^2+xy+y^2|\sqrt{(x^2+xy+y^2)^2-1}}$

13. $\frac{dz}{dt} = -1$

14. $\frac{dT}{dp} = 4p(p^2 - 1) + 2(p^3 - p)(3p^2 - 1)$

15. a. $dz = \frac{ydx - xdy}{x^2 + y^2}$ b. $du = \left(xz + \frac{x}{z}\right)^y \left[\ln\left(xz + \frac{x}{z}\right) dy + \frac{y}{xz + \frac{x}{z}} \left(\left(z + \frac{1}{z}\right) dx + x \left(1 - \frac{1}{z^2}\right) dz \right) \right]$

16. $r^2 \sin \theta$

17. -5

1. Expand $x^2y + 3y - 2$ in the power of $x - 1$ and $y + 2$ using Taylor's series.
2. Expand $e^x \cos y$ about $(0, \frac{\pi}{2})$ up to the third term using Taylor's series.
3. Show that the expansion of $\sin(xy)$ in the power of $(x - 1)$ and $(y - \frac{\pi}{2})$ up to and including second degree term is $1 - \frac{1}{8}\pi^2(x - 1)^2 - \frac{1}{2}\pi(x - 1)(y - \frac{\pi}{2}) - \frac{1}{2}(y - \frac{\pi}{2})^2$.
4. A robot moves with very small angles x and y . Its motion is represented by $f(x, y) = \cos x \cos y$. Find the Taylor expansion near $(0, 0)$.
5. A probability model in data science is defined by $f(x, y) = \sqrt{1 + x + y}$. Expand the function about $(0, 0)$ up to third-order terms.
6. Find the maxima and minima of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$.
7. Show that the function $(y - x)^4 + (x - 2)^4$ has a minimum value at $(2, 2)$.
8. Show that the function $f(x, y) = x^2 - 3xy^2 + 2y^4$ has neither a maximum nor a minimum value at the origin.
9. A data center adjusts two system parameters x and y to reduce power consumption given by $f(x, y) = x^2 + 2y^2 - 8x - 4y$. Find the minimum power consumption.
10. A game engine optimizes graphics performance using $f(x, y) = x^3 + y^3 - 3xy$. Find the stationary points and test for maxima or minima.
11. Find the smallest and largest value of $2x - y$ on the curve $x - \sin y = 0, 0 \leq y \leq 2\pi$.
12. Find the shortest distance between the line $y = 10 - 2x$ and the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$.
13. Find the extreme values of $f(x, y) = 2x + 3y + z$ such that $x^2 + y^2 = 5$ and $x + z = 1$.
14. A data center wants to minimize energy consumption $f(x, y) = x^2 + y^2$ subject to network capacity $x + 2y = 20$. Apply Lagrange multipliers to find the minimum.
15. A game developer wants to maximize graphics quality $f(x, y) = x^2y$ subject to $x + y = 12$. Find the optimal values using Lagrange multipliers.

Answers:

1. $-10 - 4(x - 1) + 4(y + 2) - 2(x - 1)^2 + 2(x - 1)(y + 2) + (x - 1)^2(y + 2)$
2. $-\left(y - \frac{\pi}{2}\right) - x\left(y - \frac{\pi}{2}\right) - \frac{1}{2}x^2\left(y - \frac{\pi}{2}\right) + \frac{1}{6}\left(y - \frac{\pi}{2}\right)^3 + \dots$
4. $f(x, y) = 1 - \frac{1}{2}(x^2 + y^2) + \frac{1}{24}(x^4 + 6x^2y^2 + y^4) + \dots$
5. $f(x, y) = 1 + \frac{x+y}{2} - \frac{1}{2}(x^2 + 2xy + y^2) + \frac{1}{16}(x^3 + 3x^2y + 3xy^2 + y^3) + \dots$
6. Local minima at (1,2) and local maxima at (-1,-2).
9. Minimum power consumption occurs at $(x, y) = (4, 1)$ and minimum value is -18 .
10. Stationary points are $(0, 0), (1, 1)$.
11. Smallest value $= -\sqrt{3} - \frac{5\pi}{3}$ and largest value $= \sqrt{3} - \frac{\pi}{3}$.
12. $\sqrt{5}$.
13. Maximum value is $1 + 5\sqrt{2}$ and minimum value is $1 - 5\sqrt{2}$.
14. Minimum energy consumption is 40.
15. The optimal values that maximize the graphics quality function are $(x = 8)$ and $(y = 4)$, yielding a maximum quality value of $f(8, 4) = 256$.

Tutorial Sheet 3 (Double Integrals, Change of Order, Change of Variable, Beta-Gamma Functions)

1. Sketch the region of integration and evaluate the integrals:

a. $\int_{y=1}^{\ln 8} \int_{x=0}^{\ln y} e^{x+y} dx dy$

c. $\int_{x=0}^{\pi} \int_{y=0}^{\sin x} y dy dx$

b. $\int_{y=0}^2 \int_{x=0}^{y^2} e^{\frac{x}{y}} dx dy$

d. $\int_{y=-1}^0 \int_{x=-1}^1 (x+y+1) dx dy$

2. Evaluate $\int \int_S xy - y^2 dx dy$ where S is the triangle with vertices $(0,0)$, $(10,1)$ and $(1,1)$.

3. Evaluate $\int \int_R y dx dy$, where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$.

4. Change the order of integration and evaluate the following:

a. $\int_{x=0}^{\infty} \int_{y=x}^{\infty} \frac{e^{-y}}{y} dy dx$

c. $\int_{x=0}^{4a} \int_{y=\frac{x^2}{4a}}^{2\sqrt{ax}} dy dx$

b. $\int_{x=0}^{\infty} \int_{y=0}^x x e^{\frac{-x^2}{y}} dy dx$

d. $\int_{x=0}^{\sqrt{\pi}} \int_{y=x}^{\sqrt{\pi}} \cos y^2 dy dx$

5. Sketch the region of integration and evaluate $\int \int_R (y - 2x^2) dx dy$ where R is the region inside the square $|x| + |y| = 1$.

6. Evaluate $\int \int_R e^{x^2+y^2} dy dx$ where R is the semicircle region bounded by the x -axis and the curve $y = \sqrt{1-x^2}$.

7. Evaluate the following by changing into polar coordinates:

a. $\int_{x=2}^2 \int_{y=0}^{\sqrt{2x-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx$

b. $\int_{y=0}^2 \int_{x=0}^{\sqrt{2^2-y^2}} (x^2 + y^2) dy dx$

8. Express the following integrals as β -function:

a. $\int_0^1 x^4 (1 - \sqrt{x})^5 dx$

c. $\int_0^1 \frac{x^2}{\sqrt{1-x^5}} dx$

b. $\int_0^1 x^2 (1 - x^3)^{\frac{3}{2}} dx$

9. Evaluate the following integrals using an appropriate substitution:

a. $\int_0^{\infty} \sqrt{x} e^{-\sqrt[3]{x}} dx$

b. $\int_0^1 \frac{1}{\sqrt{-\log x}} dx$

10. Express the following integrals in terms of the Beta or Gamma function:

a. $\int_0^{\frac{\pi}{2}} \sin^3 x \sin^{\frac{5}{2}} x dx$

c. $\int_0^1 x^2 (1 - x^3)^{\frac{3}{2}} dx$

b. $\int_0^{\frac{\pi}{2}} \sin^5 x dx$

11. Prove that

$$\int_0^{\frac{\pi}{2}} \sqrt{\sin \theta} d\theta \cdot \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin \theta}} d\theta = \pi$$

12. Show that

$$\int_0^{\frac{\pi}{2}} \sqrt{\tan x} dx = \frac{1}{2} \Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right)$$

Answers:

1. a. $8 \ln 8 - 16 + e$ b. $e^2 - 1$ c. $\frac{\pi}{4}$ d. 1

2. 6

3. $\frac{48}{5}$

4. a. 1 b. $\frac{16}{3} a^2$ c. $\frac{1}{2}$ d. 0

5. $\frac{-2}{3}$

6. $\frac{\pi(e-1)}{2}$

7. a. $\frac{4}{3}$ b. 2π

8. a. $2\beta(10,6)$ b. $\frac{1}{3}\beta(1, \frac{5}{2})$ c. $\frac{1}{5}\beta(\frac{3}{5}, \frac{1}{2})$

9. a. $\frac{315}{16}\sqrt{\pi}$ b. $\sqrt{\pi}$

10. a. $\frac{8}{77}$ b. $\frac{8}{15}$ c. $\frac{\sqrt{\pi}}{2}$

1. Evaluate $\int \int_R (x+y)^2 dx dy$ where R is a parallelogram in the x, y -plane with vertices $(1,0), (3,1), (2,2), (0,1)$ using the transformation $u = x + y$ and $v = x - 2y$.
2. Evaluate $\int \int_R (x-y) \sin(x+y) dx dy$ over the tilted square region R with corners at $(0,0), (\pi,0), (\frac{\pi}{2}, -\frac{\pi}{2}), (\frac{\pi}{2}, \frac{\pi}{2})$ by doing a change of variables $u = x + y$ and $v = x - y$.
3. Find the area bounded by the parabola $y^2 = 4 - x$ and $y^2 = 4 - 4x$ as a double integral.
4. Find the area enclosed by the curves $y = 4x - x^2$ and $x = y$.
5. Find the volume of the region that lies under the paraboloid $z = x^2 + y^2$ and above the triangle enclosed by the lines $x = y, x = 0$ and $x + y = 2$ in xy -plane.
6. An image has pixel intensity given by the continuous function $f(x, y) = x^2 + y^2$ over the square region $D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}$. Compute the total brightness $B = \int \int_D (x^2 + y^2) dA$.
7. A two-dimensional signal has power density $P(x, y) = e^{-(x^2 + y^2)}$. The total energy of the signal over a region D is defined as $E = \int \int_R P(x, y) dA$. Evaluate the total energy E when the region D is the unit disk $D = \{(x, y) | x^2 + y^2 \leq 1\}$.
8. The upper hemisphere of radius 1 is described by $z = \sqrt{1 - x^2 - y^2}$ for (x, y) lying in the unit disk $D: x^2 + y^2 \leq 1$. Compute the surface area:

$$A = \int \int_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA.$$

9. A robot's efficiency is given by $E(x, y) = 1 + xy$ over the rectangular region $R = \{(x, y) | 0 \leq x \leq 2, 0 \leq y \leq 3\}$. Compute the total efficiency:

$$T = \int \int_R (1 + xy) dA.$$

10. A rectangular CPU chip extends over the region $0 \leq x \leq 2$ cm and $0 \leq y \leq 3$ cm. The temperature distribution ($T(x, y)$ in $^{\circ}\text{C}$.) across the chip is modeled by

$$T(x, y) = 50 + 10x + 5y$$

- a. Compute the total heat measure over the chip, defined as

$$Q = \int \int_R T(x, y) dA, \text{ where } R = \{(x, y) | 0 \leq x \leq 2, 0 \leq y \leq 3\}.$$

- b. Using this result, compute the average temperature T_{avg} over the chip.

Answers:

1. 21

2. $\frac{\pi^2}{2}$

3. 8

4. 4.5

5. $\frac{4}{3}$

6. $B = \frac{2}{3}$

7. $E = \pi(1 - e^{-1})$

8. $A = 2\pi$

9. $T = 15$

10. a. $Q = 405^\circ C.Cm^2$ b. $67.5^\circ C$